

Shreya Ann Sebastian

Data Science and Machine Learning | MSc Computer Science, TU Delft

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Profile

MSc Computer Science (TU Delft, Artificial Intelligence track) and BSc Computer Science (Leiden, bioinformatics track), with a focus on data science and machine learning. MSc thesis on the social inclusion of children with hearing loss in an inclusive preschool, combining UWB positioning and LENA audio data with unsupervised methods (Gaussian mixture models, Dominant Sets clustering) and per-context linear mixed models. Current project is Tech Jobs Radar, a public job board that crawls employer career sites and job boards and extracts structured fields from each posting. Other work covers deep learning, computer vision, natural language processing and large language models.

Education

MSc Computer Science

2023 - 2026

Delft University of Technology

Artificial Intelligence Technology Track

Thesis: Automated Discovery of Latent Activity Contexts from Multimodal Sensing for Peer-Interaction Analysis in Preschoolers with Hearing Loss.

- Built an unsupervised pipeline fusing UWB spatial tracking with LENA audio to study the social inclusion of children with hearing loss in an inclusive preschool.
- Recovered six interpretable activity contexts with a Gaussian Mixture Model over acoustic and spatial features, and detected spatial groups with Dominant Sets clustering on proximity and body orientation.
- Fitted per-context linear mixed models comparing children with and without hearing loss, revealing inclusion effects that did not appear in the pooled analysis.

BSc Computer Science

2020 - 2023

Leiden University

Bioinformatics Track

Thesis: Enhancing classification accuracy of *Urticaceae* pollen using 3D pollen stack images in CNNs, as opposed to 2D projections of the same image data.

- Compared 3D CNNs on raw Z-stacks against 2D-projection baselines for classifying *Parietaria* vs. *Urtica* pollen, implementing a custom ConvNet3D and torchvision `r3d_18` ResNet3D in PyTorch on 600 variable-size stacks.
- Tuned augmentation and hyperparameters and compared whole-stack against sub-stack training; the best model, a ResNet3D trained from scratch on full stacks, reached 95% test accuracy.

International Baccalaureate Diploma

International School Eindhoven

HL: Biology, Chemistry, Dutch B | SL: Environmental Systems & Societies, English A (Language & Literature), Mathematics

Projects

Tech Jobs Radar

techjobsradar.nl [Repo](#)

Personal project, deployed

Built, deployed and run a public job board that reads postings from employer career sites and job boards, classifies tech roles and extracts structured fields from each posting. Runs on Kubernetes on AWS with automated container builds, monitoring and alerting, and nightly database backups.

Tools: Python, FastAPI, Postgres, Kubernetes, AWS

RetroSTC-Net: Retroactive Consolidation for Neural Networks

[Repo](#)

Personal research project, open source (MIT)

Implemented event-gated, retroactive weight consolidation in backprop-trained networks, inspired by synaptic tagging and capture, with a consolidation log that records which event retained which piece of learning. Evaluated on Split-MNIST over repeated seeds against EWC and Synaptic Intelligence, and validated the log with a deletion faithfulness test.

Tools: Python, PyTorch

LLM-Panel Annotation Framework



Group research project, TU Delft

Co-developed a panel of several LLMs that annotates meeting transcripts against a fixed creative-thinking codebook. Compared a baseline condition with one in which each model receives the other models' previous annotations as few-shot examples, and measured inter-model agreement.

Tools: Python, OpenAI, Gemini, Mistral and Llama APIs

Deep Learning Reproduction Project



Group project, Deep Learning course, TU Delft

Reproduced a published multi-view underwater image enhancement paper. Implemented the model architecture, contributed to data preprocessing and wrote the technical blog post reporting the reproduction results.

Tools: Python, PyTorch

Routine-Learning Smart Home System



Software Architecture course, TU Delft

Built a fault-tolerant sensor data-ingestion proof of concept validated under simulated network failures and high-concurrency load.

Tools: Python, RabbitMQ, Pydantic, Docker, GitLab CI

Personal Finance Engine

Personal project

Built a local-first Python tool that matches noisy bank statement text to merchants and detects recurring payments, evaluated on synthetic labelled data with a held-out test set.

Tools: Python, DuckDB, RapidFuzz

Coursework

MSc (TU Delft): Deep Learning, Machine Learning, Reinforcement Learning, Computer Vision, Natural Language Processing, Pattern Recognition, Socio-Cognitive Engineering, Social Signal Processing

BSc (Leiden): Algorithms & Data Structures, Probability & Statistics, Bioinformatics, Introduction to Artificial Intelligence, Data Mining, Human-Computer Interaction

Skills

Programming: Python, SQL, JavaScript, Java, C++, object-oriented programming

Data Science: Data analysis, statistical analysis, hypothesis testing (paired t-tests, Shapiro-Wilk), regression and linear mixed-effects models, feature engineering, clustering, experiment design, error analysis; pandas, NumPy, scikit-learn

Machine Learning: Supervised and unsupervised learning, classification, deep learning (CNNs), Gaussian mixture models, continual learning (EWC, SI), model evaluation (precision, recall, F1, held-out test sets), ablations over repeated seeds, explainability; PyTorch, TensorFlow, OpenCV

LLMs & NLP: Prompt engineering, few-shot prompting, structured JSON outputs, information extraction, LLM evaluation, inter-annotator agreement; OpenAI, Gemini, Mistral and Llama APIs

Data Engineering: ETL and data pipelines, SQLAlchemy, PostgreSQL, SQLite, DuckDB, Redis queues, RabbitMQ, web crawling (rate limits, robots.txt), deduplication, fuzzy text matching

Software & MLOps: REST APIs (FastAPI), Pydantic, Docker, Kubernetes (k3s, Helm), Terraform, AWS (EC2, S3, IAM), CI pipelines (GitHub Actions, GitLab CI), golden evaluation sets and regression tests in CI, unit testing (pytest), Prometheus, Cloudflare, Git

Ways of Working: Agile, Scrum, version control (Git), team projects, technical writing, presenting results to domain experts

Areas: Data Science, Artificial Intelligence (AI), Machine Learning (ML), Deep Learning, Computer Vision, Natural Language Processing (NLP), LLM Evaluation, Continual Learning, Multimodal Sensing, Statistical Modelling

Extracurriculars & Languages

TU Delft: DSBV Punch basketball | **Int. School Eindhoven:** Varsity Basketball, Student Council, Science Club

English: native | **Dutch:** conversational